

Clustering of countries based on multi-domain indicators from World Bank Data

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Abstract—Grouping world countries based on different socio-economic and environmental characteristics not only considers the more reliable categorization but also simplifies the decision-making process to support sustainable development goals. Application of unsupervised learning algorithms for clustering of countries based on various indicators gives more comprehensive and reasonable grouping by eliminating the limitations of previous studies focused on single-domain analysis or restricted to regional analysis. For this purpose, a dataset containing 47 global indicators across 267 countries over a 23-year period (2000-2022) were extracted from World Bank Open Data. Indicators used for countries' clustering involves data representing land use, population trends, CO₂ emissions and energy production/consumption. The first part of the experimental work was focused on data collection and preprocessing, the second phase involved application and comparison of several clustering methods for grouping of the countries. The ability to obtain intra-cluster and inter-cluster comparisons based on global development indicators using clusters derived from experiments offers a great advantage over policymaking and research by identification of transferable strategies for growth, sustainability and reform.

Keywords—unsupervised learning clustering, World Bank indicators, Global development patterns, Multi-domain analysis

I. INTRODUCTION (HEADING 1)

Comparing countries across multiple development indicators is essential for shaping informed policies in economics, energy, and sustainability. Traditional regional or single-indicator analyses often overlook key differences among nations. With the availability of large-scale datasets like World Bank Open Data, it is now possible to perform high-dimensional clustering to uncover groups of countries with similar socio-economic and environmental characteristics. Unsupervised learning methods, particularly clustering algorithms, allow researchers to identify hidden patterns in complex datasets without prior assumptions. By clustering countries using a broad set of indicators—spanning land use, population, energy, and emissions—this study seeks to reveal new groupings that go beyond geographic or income-based classifications. In this work, we use 47 indicators covering 266 countries over 23 years (2000–2022). The main goals are to preprocess this high-dimensional dataset and apply clustering techniques, including principal component analysis (PCA) and K-means, to identify meaningful global development patterns.

II. METHODOLOGY

A. Data pre-processing

The dataset for this study was obtained from the World Bank Open Data platform [1], comprising 266 countries and regions with 1128 columns of numerical indicators spanning the years 2000–2022. Each row represents a unique country or geographical unit, and each column corresponds to a specific indicator-year pair or aggregated indicator. To ensure analytical rigor, initial data diagnostics were conducted using the pandas library in Python [8]. The dataset contained primarily floating-point numerical values, with the exception of a single categorical Country column. Data completeness was evaluated by computing the proportion of missing values in each column. A total of 253 columns (out of 1129, including the country name) were found to be entirely empty and were thus removed from further analysis, reducing the dataset to 876 columns with at least partial data coverage. Descriptive statistics for missing values revealed substantial heterogeneity, with a mean missing rate of 37.01% and a median of 23.68%. Some indicators, such as “Land below 5m elevation (% of total land)” and “Urban land area (sq. km),” exhibited over 90% missingness, while others, such as “Total population,” were complete. All columns containing 100% missing values were dropped, and for the remaining variables, missing values were imputed using either the median or the mean, depending on the distribution of each feature. Specifically, median imputation was applied to features with pronounced skewness (absolute skewness greater than 0.5), while mean imputation was used for variables exhibiting approximately symmetric, near-normal distributions [3], [8]. After imputation, the dataset was fully numeric and contained no missing values.

Table I provides descriptive statistics for a subset of 17 key indicators used in the analysis, including their temporal coverage, country counts, distribution ranges, and missing data percentages.

Table I. Descriptive Summary of Selected World Bank Indicators (2000–2022)

B. Feature Scaling and Encoding

clusters are, indicated that $k=8$ produced the highest value. Thus, we proceeded with $k=8$ as the final number of clusters.

	Indicator code	Description	Years	Countries	Min	Max	Mean	Median	Missing %
1	AG.LND.AGRI.ZS	Agricultural land (% of land area)	2000–2022	266	0.44%	85.49 %	37.28%	38.47%	7.70%
2	AG.LND.ARBL.ZS	Arable land (% of land area)	2000–2022	266	0.04%	64.15 %	13.50%	9.89%	8.80%
3	AG.LND.EL5M.ZS	Land below 5m elevation (% of total land)	2000–2022	266	0%	81.70 %	5.92%	1.28%	91.80%
4	AG.LND.FRST.ZS	Forest area (% of land area)	2000–2022	266	0%	95.58 %	32.22%	30.75%	6.80%
5	AG.LND.PRCP.MM	Avg. annual precipitation (mm)	2000–2022	266	18.1	3,240.0	1,174.00	1,032.00	37.70%
6	AG.LND.TOTL.K2	Total land area (sq. km)	2000–2022	266	~2	129,955,200	~4.97M	196,850	6.10%
7	AG.LND.TOTL.UR.K2	Urban land area (sq. km)	2000–2022	266	0	1,907,334	67,106	2,002	91.80%
8	EG.ELC.FOSL.ZS	Fossil fuel electricity (% of total)	2000–2022	266	0%	100%	61.10%	65.82%	53.60%
9	EG.ELC.RNEW.ZS	Renewable electricity output (% of total)	2000–2022	266	0%	100%	28.12%	17.59%	33.50%
10	EG.ELC.RNWX.ZS	Renewable energy use (% of energy consumption)	2000–2022	266	0%	65.44 %	3.16%	0.71%	53.60%
11	EG.IMP.CONS.ZS	Energy imports, net (% of energy use)	2000–2022	266	-259%	101.00%	0.24%	-0.16%	6.60%
12	EG.USE.ELEC.KH.PC	Electric power consumption (kWh per capita)	2000–2022	266	0.3	53,250.00	3,596.70	1,392.70	11.20%
13	EN.ATM.CO2E.KD.GD	CO ₂ intensity (kg per 2017 PPP \$ of GDP)	2000–2022	266	0.0002	5.69	0.43	0.33	13.20%
14	EN.ATM.CO2E.PC	CO ₂ emissions (metric tons per capita)	2000–2022	266	0	74.3	3.99	2.2	12.30%
15	EN.ATM.PM25.MC.M3	PM2.5 air pollution, mean annual exposure ($\mu\text{g}/\text{m}^3$)	2000–2022	266	0.1	103.3	28.71	24.49	6.80%
16	EN.ATM.PM25.SG.MS.ZS	Population exposed to > WHO PM2.5 guideline (%)	2000–2022	266	0%	100%	83.29%	93.85%	7.60%
17	EN.ATM.PM25.SG.MS	Population exposed to > WHO PM2.5 guideline (millions)	2000–2022	266	0	1.41B	54.59M	1.15M	7.50%

To prepare for clustering, all numerical columns were standardized using z-score normalization (mean = 0, standard deviation = 1) via the StandardScaler in scikit-learn [8]. This transformation ensured uniform scaling across indicators with diverse units (e.g., percentages, kilotons, per-capita rates) and prevented variables with large absolute values from dominating distance computations. The categorical Country column was one-hot encoded, producing binary indicator columns for each country (minus one to avoid multicollinearity) [8]. The final dataset at this stage contained 266 rows and 1135 columns, all numeric.

C. Dimensionality Reduction

Due to the high dimensionality introduced by one-hot encoding and the large number of original indicators, Principal Component Analysis (PCA) [3] was applied to reduce the feature space while retaining 95% of the dataset's variance. This step improved computational efficiency and enhanced the quality of clustering by eliminating redundant or noise features.

III. PREPARE YOUR PAPER BEFORE STYLING

A. Clustering Algorithm and Optimal k Selection

Following data preprocessing and PCA dimensionality reduction, K-means clustering was applied to identify latent groupings among the 266 countries based on their environmental, demographic, and economic indicators. To determine the optimal number of clusters (k), silhouette analysis was conducted across k values ranging from 2 to 10. The silhouette score, which evaluates how well-separated

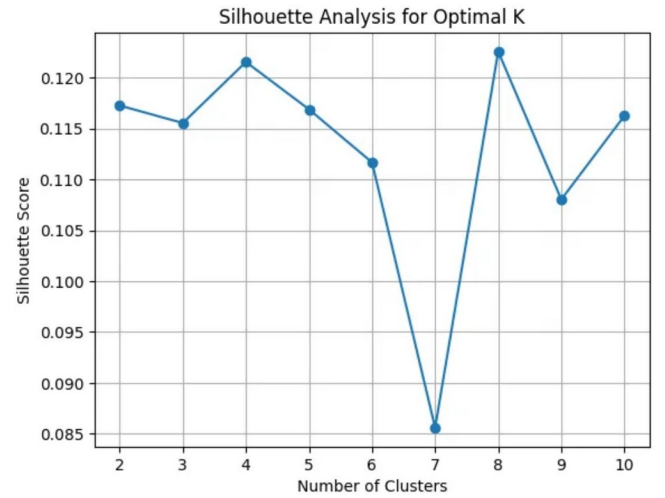


TABLE I. DESCRIPTION OF SOME INDICATORS USED IN PROCESSING

B. Cluster Characteristics

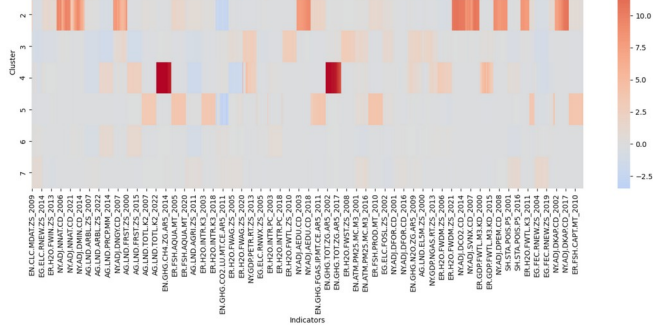
After applying K-means with $k=8$, each country was assigned a cluster label. The mean values of key indicators within each cluster were computed to characterize group-level traits. For instance:

- Cluster 0 typically included middle-income countries with mixed energy profiles and moderate CO₂ emissions.

- Cluster 3 featured nations with strong renewable electricity output.
- Cluster 7 showed the highest adoption of renewables and lowest fossil fuel reliance.

These results suggest meaningful global development patterns and energy transition stages across clusters.

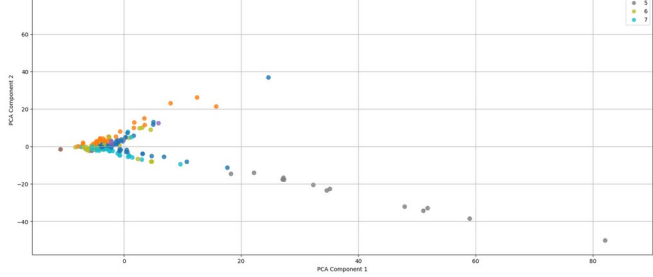
Fig. 2. Normalized indicator means for each cluster.



C. PCA Visualization

To visualize cluster separability, the high-dimensional dataset was projected into a two-dimensional space using PCA. Countries were plotted using their first two principal components and colored by cluster labels. The 2D scatter plot revealed distinct and dense groupings, confirming the validity of the clustering structure and the effectiveness of PCA in preserving inter-country differences.

Fig. 3. PCA projection of countries colored by cluster labels.

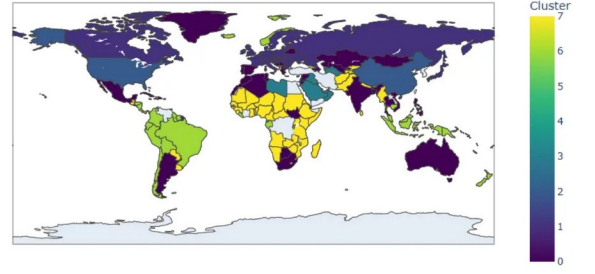


D. Geographic Distribution of Clusters

A choropleth world map was generated using ISO Alpha-3 codes and cluster labels. It revealed interesting geographical patterns:

- Cluster 0: Spread across MENA, South Asia, and Eastern Europe.
- Cluster 7: Mostly composed of developed nations with advanced sustainability policies.
- Cluster 1 and 2: Included countries with high CO_2 emissions but varied socioeconomic characteristics.

Fig. 4. Global distribution of clusters ($K=8$) using ISO country codes.



E. Azerbaijan Case Study

Azerbaijan was assigned to Cluster 0, which includes countries such as Turkey, Kazakhstan, Morocco, and India. This cluster represents emerging economies with ongoing energy transition efforts. Azerbaijan shares characteristics like fossil fuel dependence and evolving renewable energy adoption. Compared to cluster leaders like Portugal or Morocco, Azerbaijan shows room for growth in renewable energy infrastructure, environmental governance, and economic diversification.

The comparative analysis indicates that Azerbaijan can benefit by adopting best practices from cluster peers, including expanding solar/wind projects, improving environmental regulations, and fostering innovation ecosystems.

IV. DISCUSSION

A. Overview of Cluster Profiles and Interpretations

Each of the eight identified clusters reflects a unique combination of socio-economic, environmental, and energy characteristics. These profiles offer insight into how countries align in terms of development priorities, resource usage, and sustainability practices.

- Cluster 0: Transitional economies with moderate development indicators and fossil-fuel dominated energy systems. Countries here are beginning to diversify energy sources but remain reliant on hydrocarbons.
- Cluster 1: Industrialized nations with high CO_2 intensity and relatively lower adoption of renewable energy. Often exhibit large-scale manufacturing and urban density.
- Cluster 2: Countries with high forest coverage and low emissions. Often include nations with significant biodiversity and environmental reserves.
- Cluster 3: Sustainability frontrunners with high renewable electricity output, relatively low pollution, and active environmental governance.
- Cluster 4: Countries undergoing urbanization with medium-level renewable energy usage and mixed pollution indicators.
- Cluster 5: Lower-income countries with limited infrastructure and development. Characterized by low energy use and high population growth.
- Cluster 6: Middle-tier performers with improving energy metrics and moderate emissions.
- Cluster 7: High-income, low-emission nations leading in green energy adoption, smart infrastructure, and digital innovation.

These groupings reveal more nuanced global development patterns than traditional income-based or regional classifications.

B. Azerbaijan's Position within Cluster 0

Azerbaijan's inclusion in Cluster 0 positions it among countries with transitional economies—those actively navigating the shift from fossil-fuel dependence toward more sustainable energy practices. Cluster 0 includes a diverse mix of regions, from South Asia to North Africa and Eastern Europe, united by similar challenges: legacy energy infrastructure, uneven renewable deployment, and evolving regulatory frameworks.

While Azerbaijan has made progress in macroeconomic stability and regional trade, it still faces challenges in green transformation. Compared to its cluster peers like Morocco (massive solar investment), Portugal (high wind/solar penetration), or Türkiye (fast-growing manufacturing and renewables), Azerbaijan lags in key sustainability indicators. However, Azerbaijan also has unique opportunities. Its strategic geographic location, strong state capacity, and experience in large-scale energy projects (e.g., the Southern Gas Corridor) can be leveraged to scale up renewable investments and improve cross-border energy cooperation.

V. CONCLUSION

This study demonstrated how unsupervised learning techniques, specifically PCA and K-means clustering, can be used to uncover meaningful groupings of countries based on multi-domain development indicators. By clustering 266 countries using 47 indicators across environmental, energy, and demographic dimensions, we identified eight distinct clusters, each with unique sustainability characteristics and developmental trajectories.

Azerbaijan's placement in Cluster 0 highlights both its commonalities with resource-reliant transitional economies and its potential to shift toward a more sustainable, innovation-driven future. Cluster analysis not only revealed global development patterns but also facilitated peer-based benchmarking, providing actionable insights for countries aiming to emulate more sustainable models within their cluster.

The combination of clustering results and regional interpretation offers a powerful decision-support framework for governments, researchers, and policymakers to craft data-driven strategies for inclusive, green, and resilient growth.

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